

Create a C program to manage student marks for multiple departments. The program must use a **three-dimensional array** to store the data and perform the following tasks:

- **Data Structure:** The array `marks[5][3][5]` should be used to store marks for **5 departments**, with **3 students** per department, each taking **5 subjects**.
- **User Input:** The program must first prompt the user to input marks for every student in every subject across all 5 departments.
- **Total and Average Calculation:** After data entry is complete, the program should ask the user to select a specific department. For the chosen department, it must then calculate and display the **total marks** and **average marks** for each student.
- **Indexing:** The program should handle the user-friendly input (e.g., department 1, student 1) by correctly converting it to the 0-based indexing required for the C array.

Sample output :

```
PS C:\Users\STUDENT\Desktop\cprograms> gcc .\multidimarray.c
PS C:\Users\STUDENT\Desktop\cprograms> ./a.exe

=== Entering marks for Department 1 ===

Student 1:
  Subject 1 marks: 45
  Subject 2 marks: 56
  Subject 3 marks: 67
  Subject 4 marks: 54
  Subject 5 marks: 54

Student 2:
  Subject 1 marks: 45
  Subject 2 marks: 67
  Subject 3 marks: 76
  Subject 4 marks: 54
  Subject 5 marks: 45

=== Entering marks for Department 2 ===

Student 1:
  Subject 1 marks: 54
  Subject 2 marks: 54
  Subject 3 marks: 65
  Subject 4 marks: 56
  Subject 5 marks: 65

Student 2:
  Subject 1 marks: 65
  Subject 2 marks: 7
  Subject 3 marks: 54
  Subject 4 marks: 67
  Subject 5 marks: 54

Enter department number to display (1-5): 2

===== Department 2 Results =====
Student 1 -> Total: 294, Average: 58.80
Student 2 -> Total: 247, Average: 49.40
PS C:\Users\STUDENT\Desktop\cprograms> █
```

For your reference :

```
#include <stdio.h>
```

```
int main() {
    int arr[2][3][2]; // arr[x][y][z]
    int i, j, k;
    // Taking input
    printf("Enter values for a 3D array (2 x 3 x 2):\n");
    for (i = 0; i < 2; i++) { // First dimension
        for (j = 0; j < 3; j++) { // Second dimension
            for (k = 0; k < 2; k++) { // Third dimension
                printf("Enter value for arr[%d][%d][%d]: ", i, j, k);
                scanf("%d", &arr[i][j][k]);
            }
        }
    }
}
```

```
    }  
}  
  
// Displaying output  
printf("\nYou entered:\n");  
// Printing the first dimension (page index) and its contents  
for (int i = 0; i < 2; i++) {  
    printf("--- Page %d ---\n", i); // This prints the page index  
    for (int j = 0; j < 3; j++) {  
        for (int k = 0; k < 2; k++) {  
            printf("%d ", arr[i][j][k]);  
        }  
        printf("\n");  
    }  
    printf("\n");  
}  
return 0;  
}
```

Enter values for a 3D array (2 x 3 x 2):

Enter value for arr[0][0][0]: 1

Enter value for arr[0][0][1]: 2

Enter value for arr[0][1][0]: 32

Enter value for arr[0][1][1]: 43

Enter value for arr[0][2][0]: 34

Enter value for arr[0][2][1]: 34

Enter value for arr[1][0][0]: 34

Enter value for arr[1][0][1]: 43

Enter value for arr[1][1][0]: 32

Enter value for arr[1][1][1]: 32

Enter value for arr[1][2][0]: 32

Enter value for arr[1][2][1]: 32

You entered:

--- Page 0 ---

1 2

32 43

34 34

--- Page 1 ---

34 43

32 32

32 32